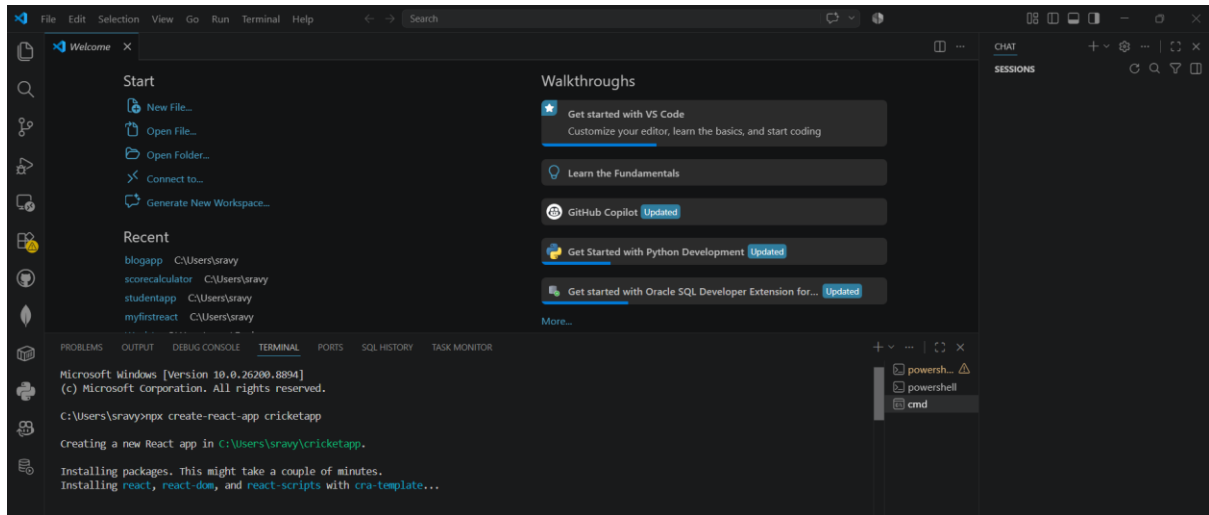


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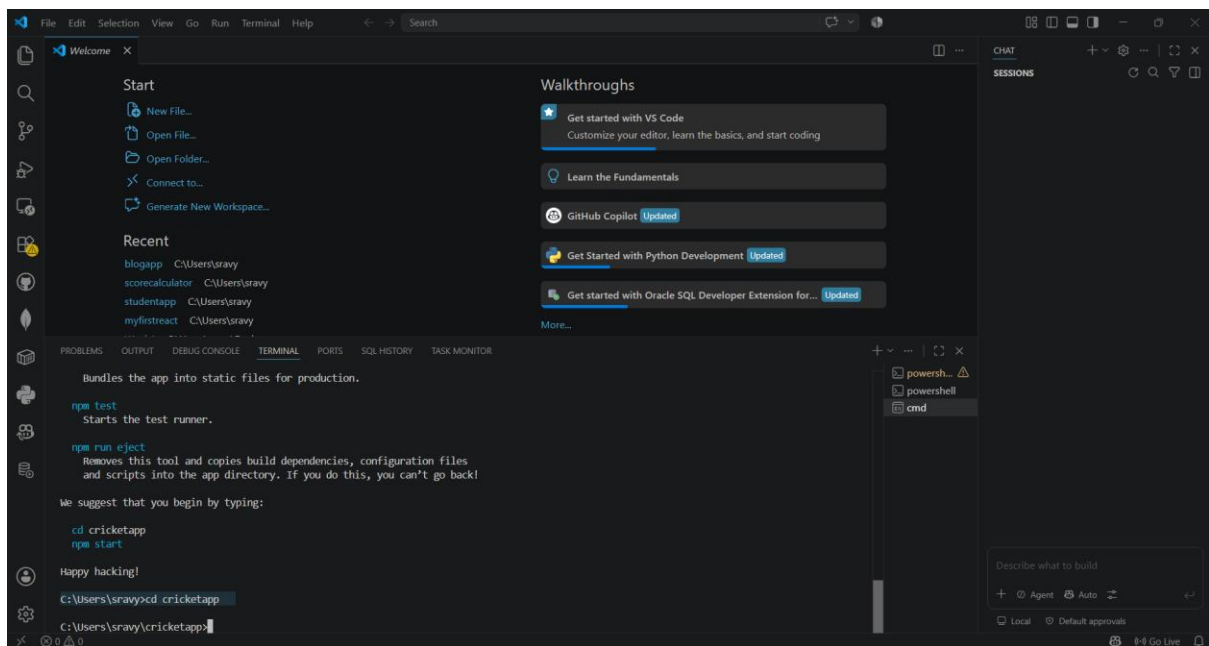
React

9. ReactJS-HOL

Step 1: Creating the React Application (cricketapp) Using Create React App

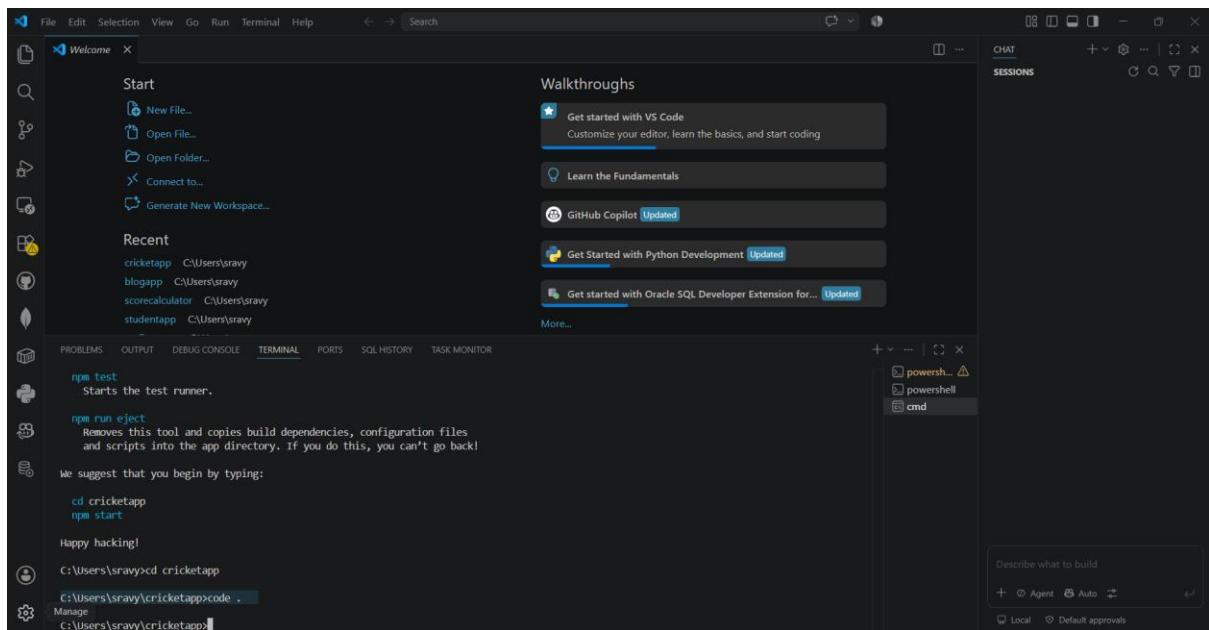


Step 2: Successfully Creating the React Project and Navigating to the Project Directory

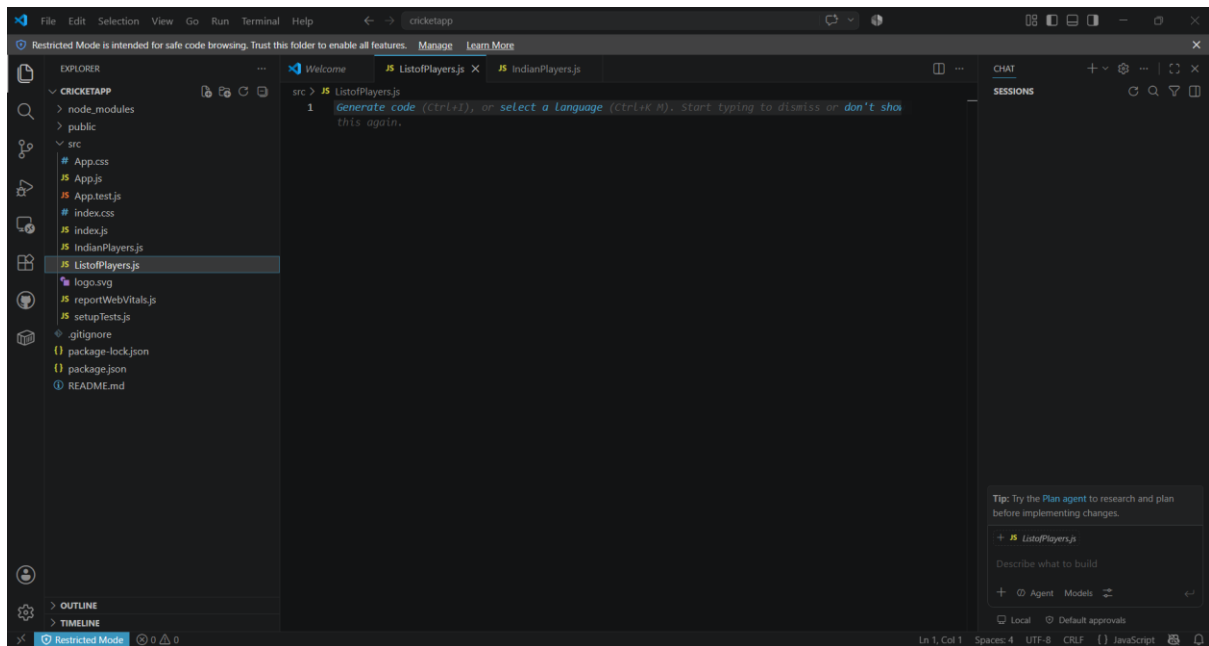


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Step 3: Opening the React Project (cricketapp) in Visual Studio Code

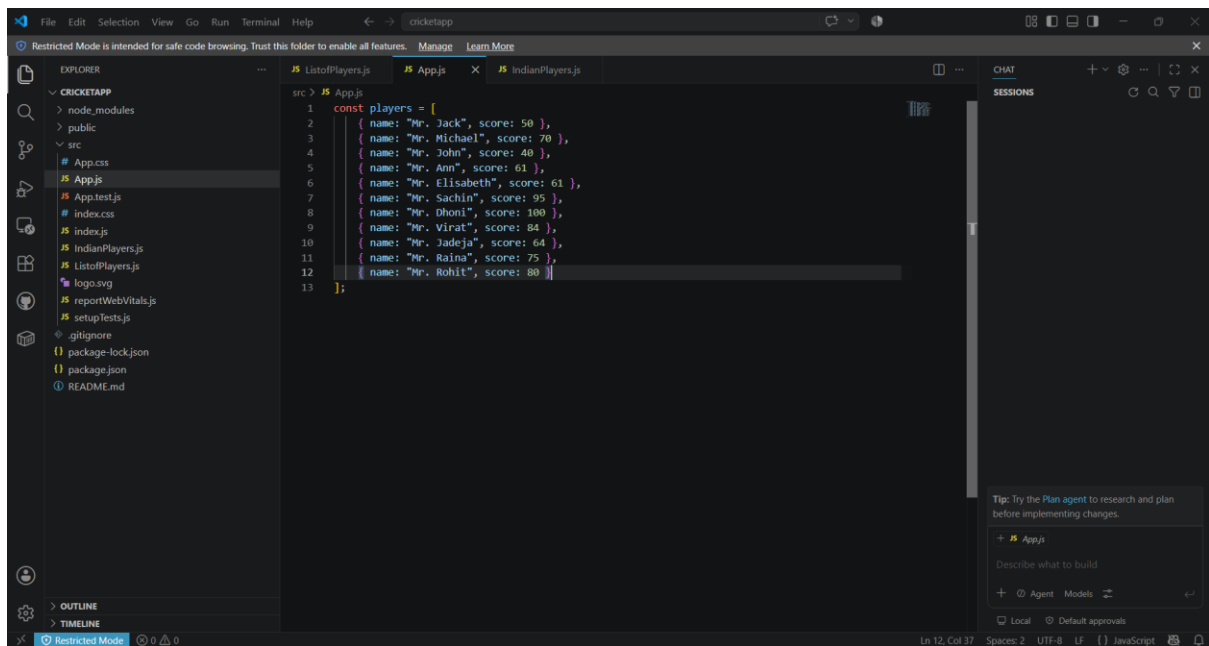


Step 4: Creating the React Components



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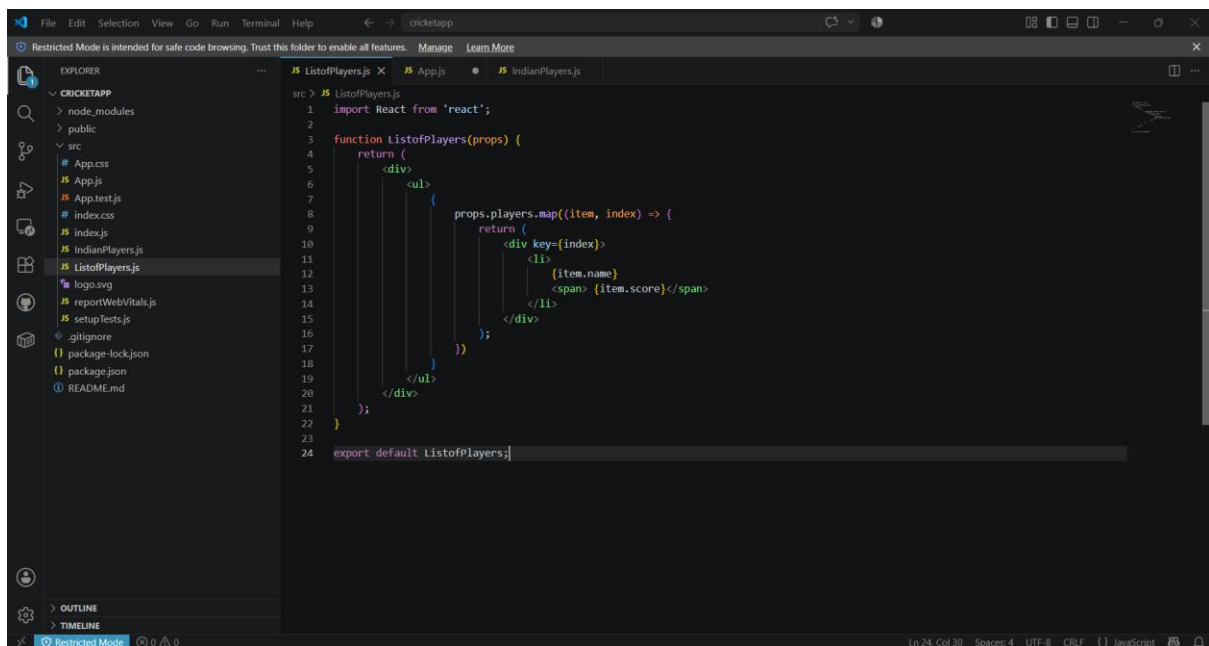
Step 5: Declaring the Players Array Using ES6



The screenshot shows the VS Code editor with the file `ListofPlayers.js` open. The code defines a constant array `players` containing 10 objects, each representing a player with a name and a score. The Explorer sidebar on the left shows the project structure for `CRICKETAPP`, including files like `App.js`, `App.test.js`, `index.css`, `index.js`, `IndianPlayers.js`, `ListofPlayers.js`, `logo.svg`, `reportWebVitals.js`, `setupTests.js`, `.gitignore`, `package-lock.json`, `package.json`, and `README.md`. The status bar at the bottom indicates the file is in `Restricted Mode`.

```
src > JS App.js
1  const players = [
2    { name: "Mr. Jack", score: 50 },
3    { name: "Mr. Michael", score: 70 },
4    { name: "Mr. John", score: 40 },
5    { name: "Mr. Ann", score: 61 },
6    { name: "Mr. Elisabeth", score: 61 },
7    { name: "Mr. Sachin", score: 95 },
8    { name: "Mr. Dhoni", score: 100 },
9    { name: "Mr. Virat", score: 84 },
10   { name: "Mr. Jadeja", score: 64 },
11   { name: "Mr. Raina", score: 75 },
12   { name: "Mr. Rohit", score: 80 }
13 ];
```

Step 6: Displaying the List of Players Using the ES6 map() Method

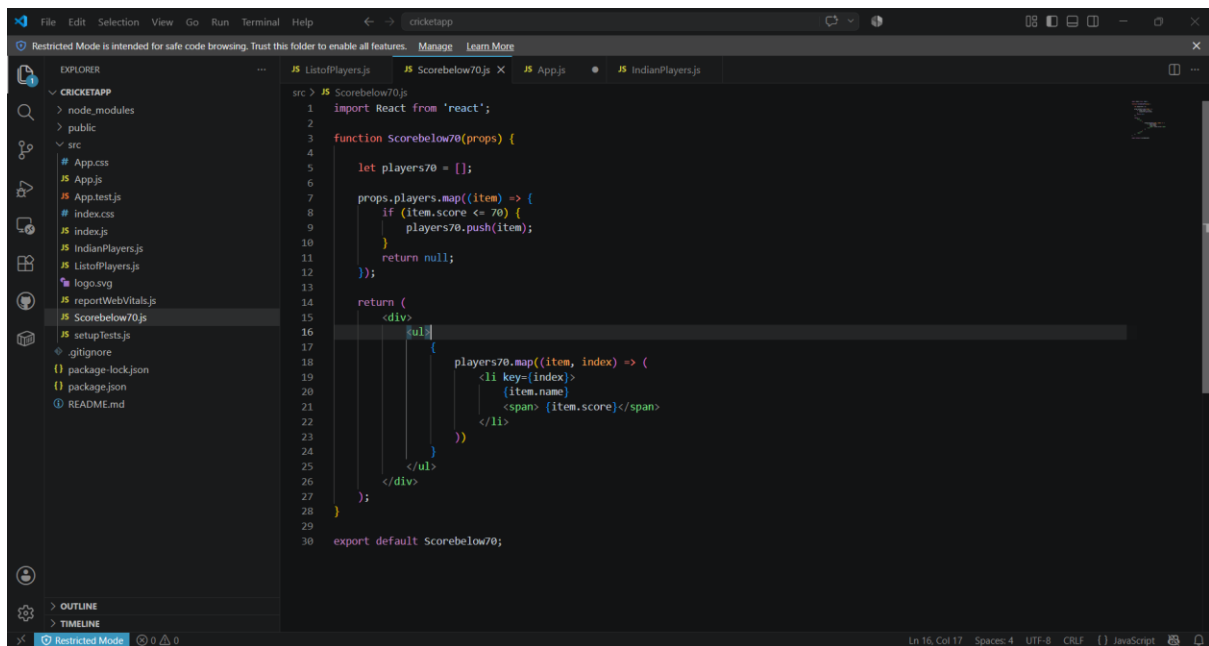


The screenshot shows the VS Code editor with the file `ListofPlayers.js` open. The code defines a function `ListofPlayers` that takes `props` as an argument and returns a JSX element. It uses the `map` method to iterate over `props.players` and render a list of players. The Explorer sidebar on the left shows the project structure for `CRICKETAPP`, including files like `App.js`, `App.test.js`, `index.css`, `index.js`, `IndianPlayers.js`, `ListofPlayers.js`, `logo.svg`, `reportWebVitals.js`, `setupTests.js`, `.gitignore`, `package-lock.json`, `package.json`, and `README.md`. The status bar at the bottom indicates the file is in `Restricted Mode`.

```
src > JS ListofPlayers.js
1  import React from 'react';
2
3  function ListofPlayers(props) {
4    return (
5      <div>
6        <ul>
7          {props.players.map((item, index) => {
8            return (
9              <div key={index}>
10                <li>
11                  {item.name}
12                  <span> {item.score}</span>
13                </li>
14              </div>
15            );
16          })}
17        </ul>
18      </div>
19    );
20  }
21
22  export default ListofPlayers;
```

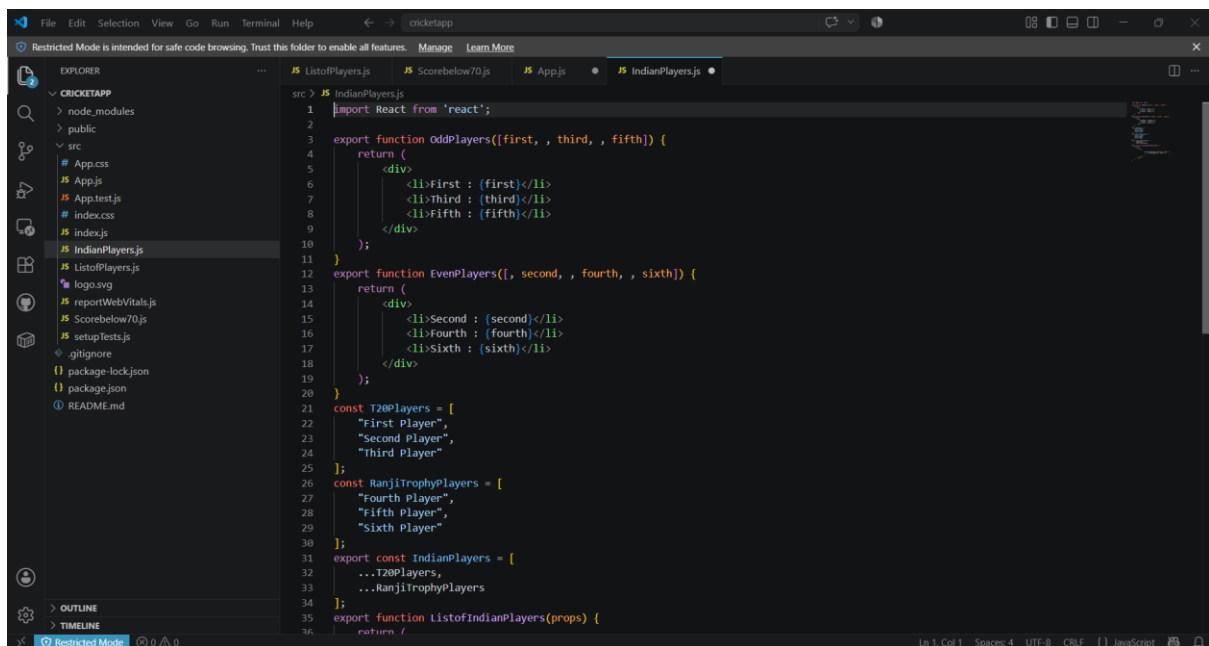
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Step 7: Filtering Players with Scores Less Than 70 Using Arrow Functions



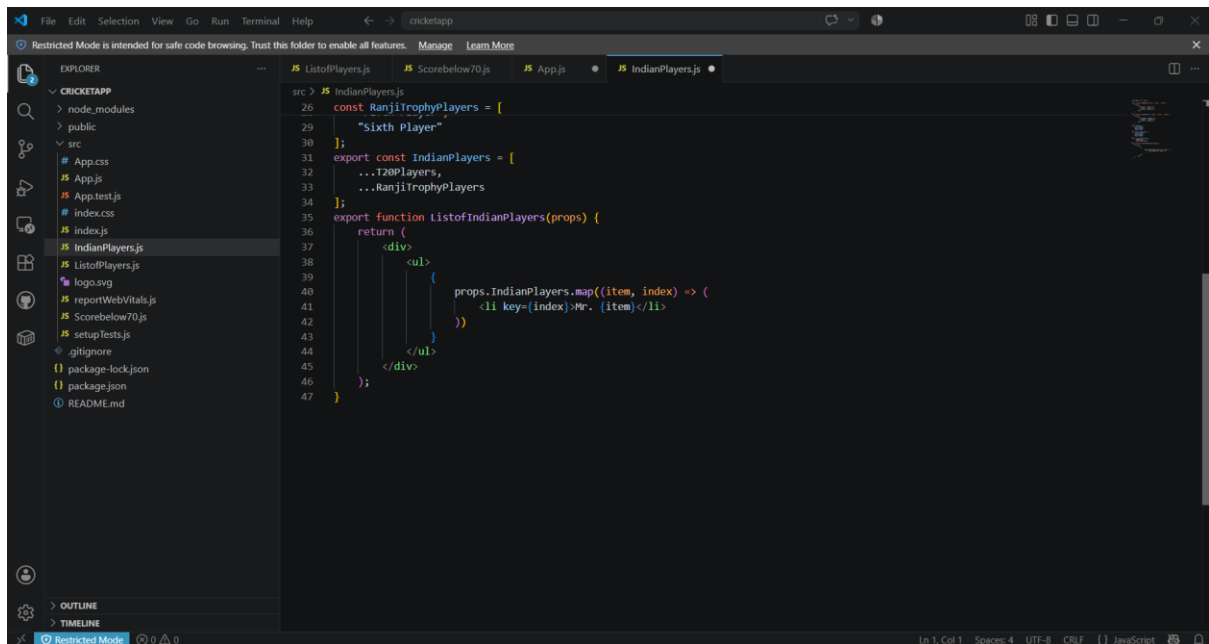
```
src > JS Scorebelow70.js
1  import React from 'react';
2
3  function Scorebelow70(props) {
4
5      let players70 = [];
6
7      props.players.map((item) => {
8          if (item.score <= 70) {
9              players70.push(item);
10         }
11         return null;
12     });
13
14     return (
15         <div>
16             <ul>
17                 {
18                     players70.map((item, index) => (
19                         <li key={index}>
20                             {item.name}
21                             <span> {item.score}</span>
22                         </li>
23                     ))
24                 }
25             </ul>
26         </div>
27     );
28 }
29
30 export default Scorebelow70;
```

Step 8: Implementing ES6 Destructuring and Merging Arrays in IndianPlayers.js

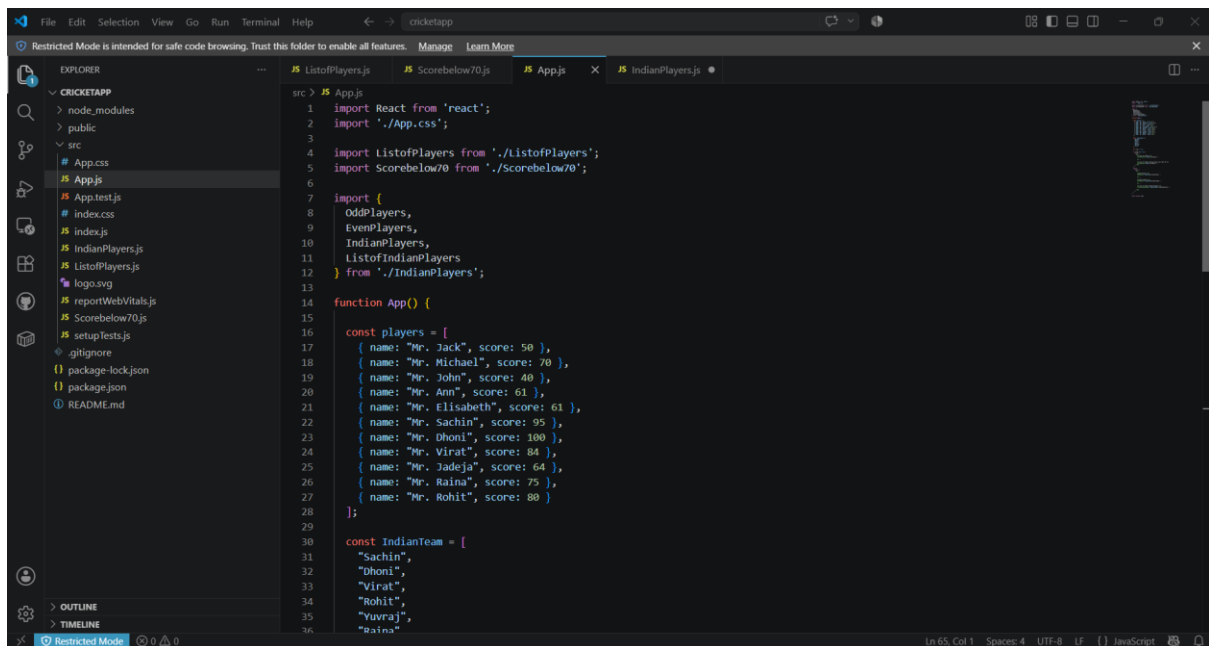


```
src > JS IndianPlayers.js
1  import React from 'react';
2
3  export function OddPlayers([first, , third, , fifth]) {
4      return (
5          <div>
6              <li>First : {first}</li>
7              <li>Third : {third}</li>
8              <li>Fifth : {fifth}</li>
9          </div>
10     );
11 }
12
13 export function EvenPlayers([, second, , fourth, , sixth]) {
14     return (
15         <div>
16             <li>Second : {second}</li>
17             <li>Fourth : {fourth}</li>
18             <li>Sixth : {sixth}</li>
19         </div>
20     );
21 }
22
23 const T20Players = [
24     "First Player",
25     "Second Player",
26     "Third Player"
27 ];
28
29 const RanjiTrophyPlayers = [
30     "Fourth Player",
31     "Fifth Player",
32     "Sixth Player"
33 ];
34
35 export const IndianPlayers = [
36     ...T20Players,
37     ...RanjiTrophyPlayers
38 ];
39
40 export function ListofIndianPlayers(props) {
41     return (
42         <div>
43             <ul>
44                 {
45                     IndianPlayers.map((player, index) => (
46                         <li key={index}>
47                             {player}
48                         </li>
49                     ))
50                 }
51             </ul>
52         </div>
53     );
54 }
```

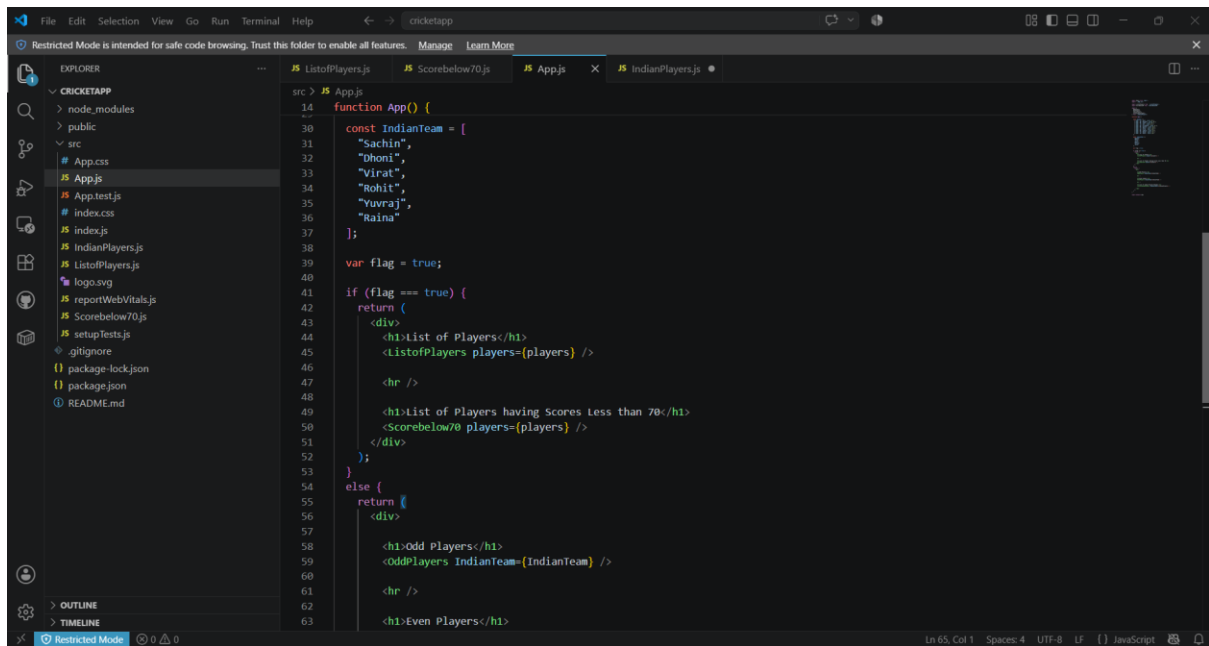
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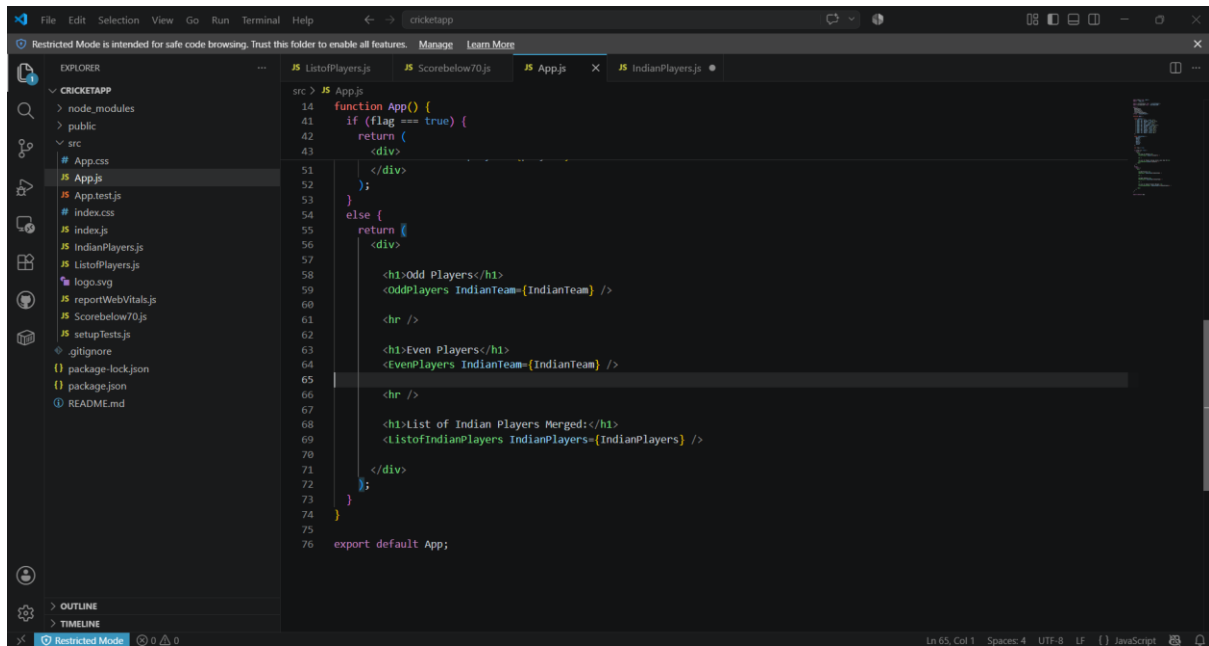
Step 9: Updating App.js to Display Components Using ES6 Features



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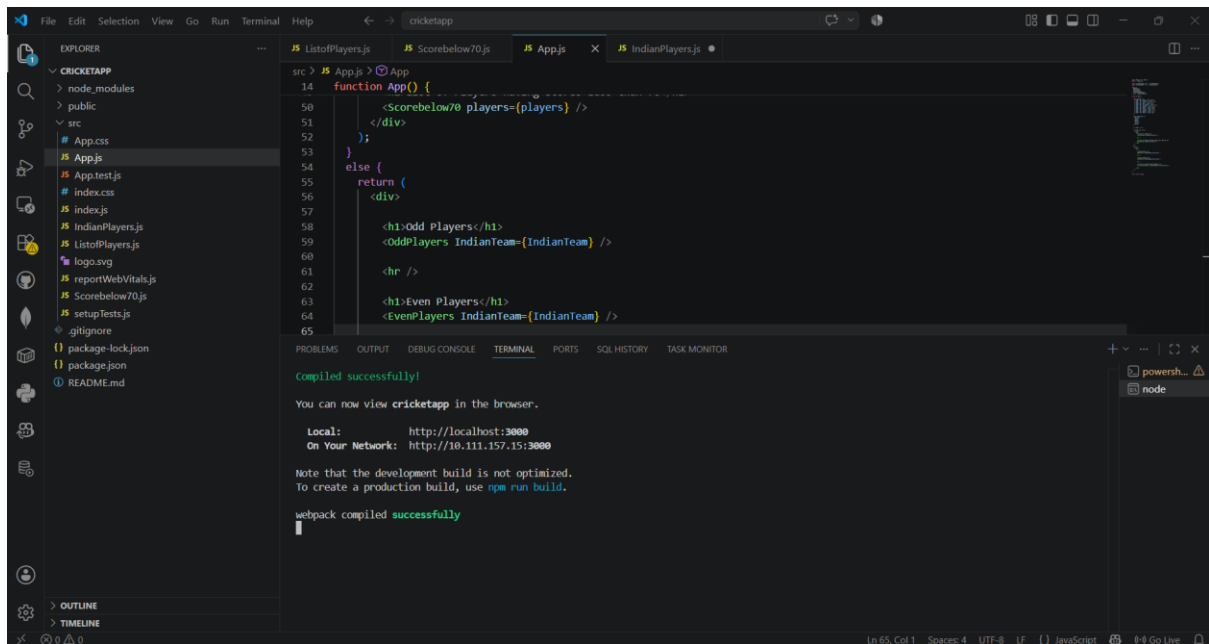
```
14 function App() {
15   const IndianTeam = [
16     "Sachin",
17     "Dhoni",
18     "Virat",
19     "Rohit",
20     "Yuvraj",
21     "Raina"
22   ];
23   var flag = true;
24   if (flag === true) {
25     return (
26       <div>
27         <h1>List of Players</h1>
28         <ListofPlayers players={players} />
29         <hr />
30         <h1>List of Players having Scores Less than 70</h1>
31         <Scorebelow70 players={players} />
32       </div>
33     );
34   }
35   else {
36     return (
37       <div>
38         <h1>Odd Players</h1>
39         <OddPlayers IndianTeam={IndianTeam} />
40         <hr />
41         <h1>Even Players</h1>
42       </div>
43     );
44   }
45 }
```



```
14 function App() {
15   if (flag === true) {
16     return (
17       <div>
18         <h1>List of Players</h1>
19         <ListofPlayers players={players} />
20         <hr />
21         <h1>List of Players having Scores Less than 70</h1>
22         <Scorebelow70 players={players} />
23       </div>
24     );
25   }
26   else {
27     return (
28       <div>
29         <h1>Odd Players</h1>
30         <OddPlayers IndianTeam={IndianTeam} />
31         <hr />
32         <h1>Even Players</h1>
33         <EvenPlayers IndianTeam={IndianTeam} />
34         <hr />
35         <h1>List of Indian Players Merged</h1>
36         <ListofIndianPlayers IndianPlayers={IndianPlayers} />
37       </div>
38     );
39   }
40 }
```

Step 10: Run the application

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The screenshot shows the Visual Studio Code editor with a project named 'cricketapp'. The Explorer panel on the left shows the file structure, including 'src' and 'public' directories. The main editor displays the 'App.js' file, which contains a function 'App()' that renders a list of players and a section for players with scores below 70. The terminal at the bottom shows the command 'npm run build' being executed, resulting in a successful build. The output includes the following text:

```
Compiled successfully!

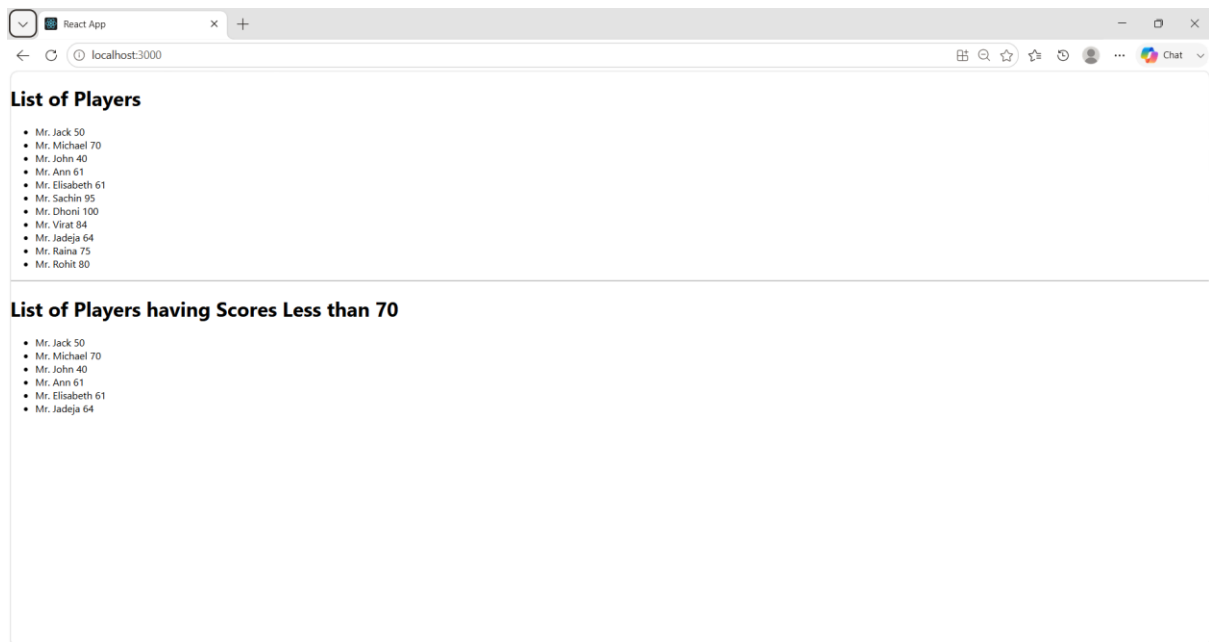
You can now view cricketapp in the browser.

Local:      http://localhost:3000
On Your Network:  http://10.111.157.15:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

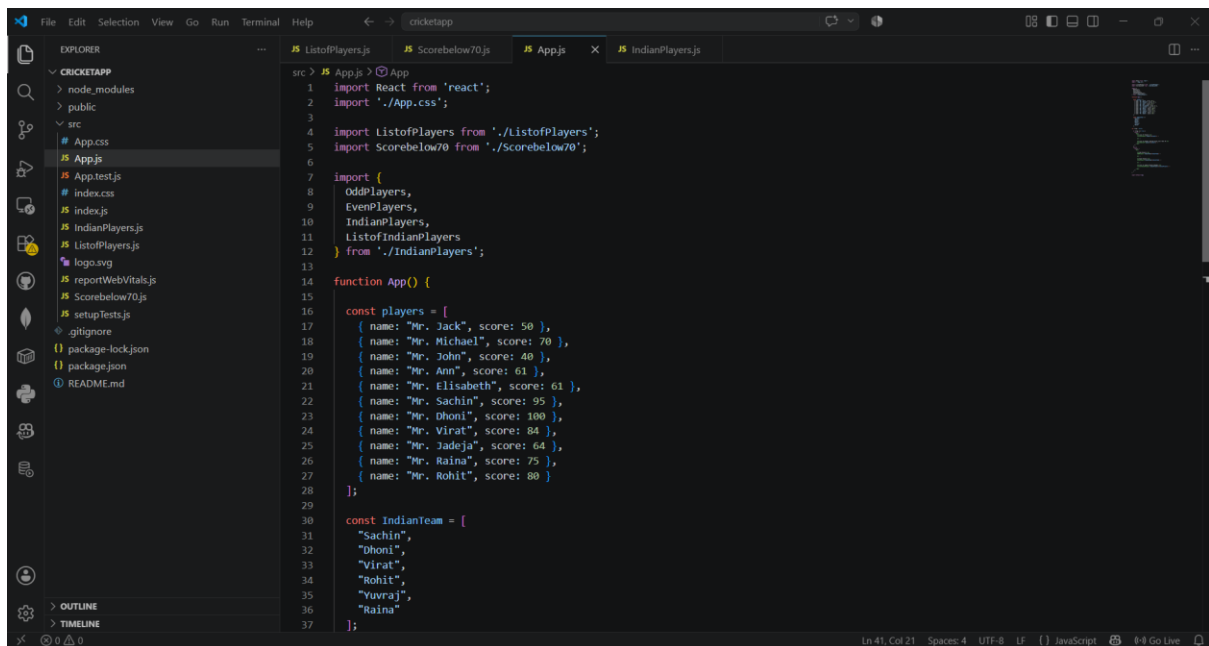
webpack compiled successfully
```

- **var flag = true**

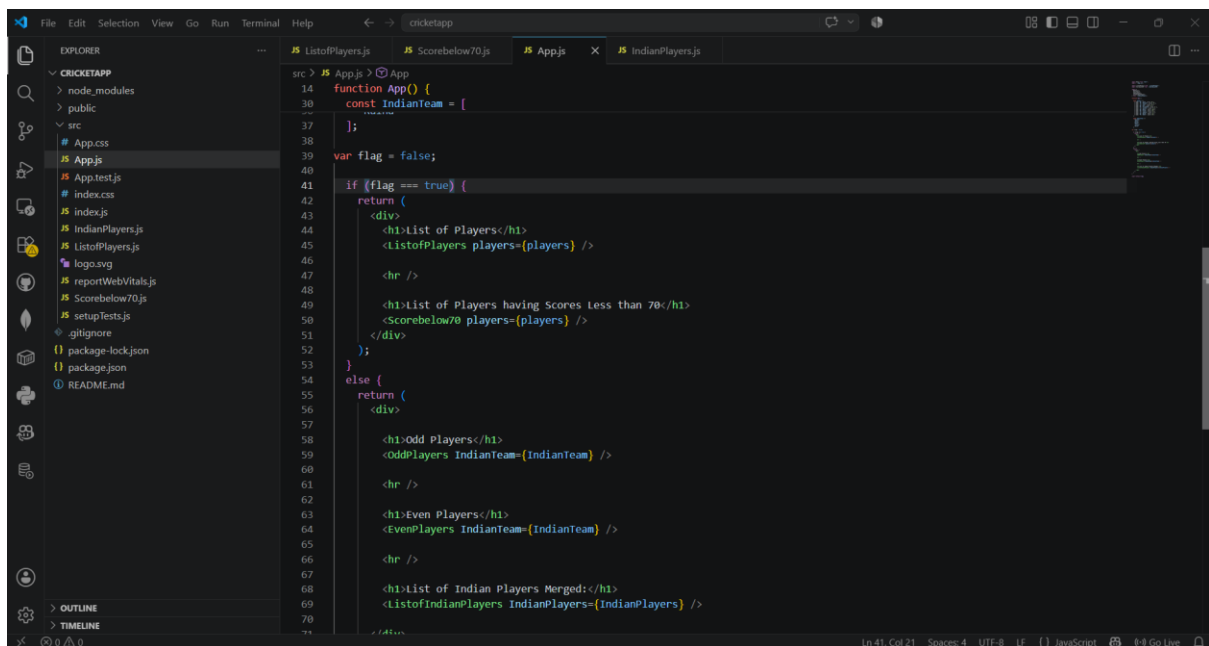


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- `var flag = false`



```
src > JS App.js > @App
1 import React from 'react';
2 import './App.css';
3
4 import ListofPlayers from './ListofPlayers';
5 import Scorebelow70 from './Scorebelow70';
6
7 import {
8   OddPlayers,
9   EvenPlayers,
10  IndianPlayers,
11  ListofIndianPlayers
12 } from './IndianPlayers';
13
14 function App() {
15
16   const players = [
17     { name: "Mr. Jack", score: 50 },
18     { name: "Mr. Michael", score: 70 },
19     { name: "Mr. John", score: 40 },
20     { name: "Mr. Ann", score: 61 },
21     { name: "Mr. Elisabeth", score: 61 },
22     { name: "Mr. Sachin", score: 95 },
23     { name: "Mr. Dhoni", score: 100 },
24     { name: "Mr. Virat", score: 84 },
25     { name: "Mr. Jadeja", score: 64 },
26     { name: "Mr. Raina", score: 75 },
27     { name: "Mr. Rohit", score: 80 }
28   ];
29
30   const IndianTeam = [
31     "Sachin",
32     "Dhoni",
33     "Virat",
34     "Rohit",
35     "Yuvraj",
36     "Raina"
37   ];
38 }
```



```
src > JS App.js > @App
14 function App() {
30   const IndianTeam = [
37   ];
38
39   var flag = false;
40
41   if (flag === true) {
42     return (
43       <div>
44         <h1>List of Players</h1>
45         <ListofPlayers players={players} />
46
47         <hr />
48
49         <h1>List of Players having Scores Less than 70</h1>
50         <Scorebelow70 players={players} />
51       </div>
52     );
53   }
54   else {
55     return (
56       <div>
57
58         <h1>Odd Players</h1>
59         <OddPlayers IndianTeam={IndianTeam} />
60
61         <hr />
62
63         <h1>Even Players</h1>
64         <EvenPlayers IndianTeam={IndianTeam} />
65
66         <hr />
67
68         <h1>List of Indian Players Merged</h1>
69         <ListofIndianPlayers IndianPlayers={IndianPlayers} />
70
71       </div>
72     );
73   }
74 }
```


COGNIZANT DIGITAL NURTURE 5.0 – JAVA FSE

```
src > JS App.js > App
14 function App() {
27   return (
56     <div>
57
58       <h1>Odd Players</h1>
59       <OddPlayers IndianTeam={IndianTeam} />
60
61       <hr />
62
63       <h1>Even Players</h1>
64       <EvenPlayers IndianTeam={IndianTeam} />
65
66       <hr />
67
68       <h1>List of Indian Players Merged:</h1>
69       <ListofIndianPlayers IndianPlayers={IndianPlayers} />
70
71     </div>
72   );
73 };
74
75 export default App;
```

